



PARALLEL SAMPLE SORT

a hybrid Open MPI + OMP implementation
and Native and Containerized benchmarking

PARALLEL SAMPLE SORT

ingredients



N unsorted keys (64 unsigned integers)



P processors (Open MPI))



T threads per process (OMP)

PARALLEL SAMPLE SORT

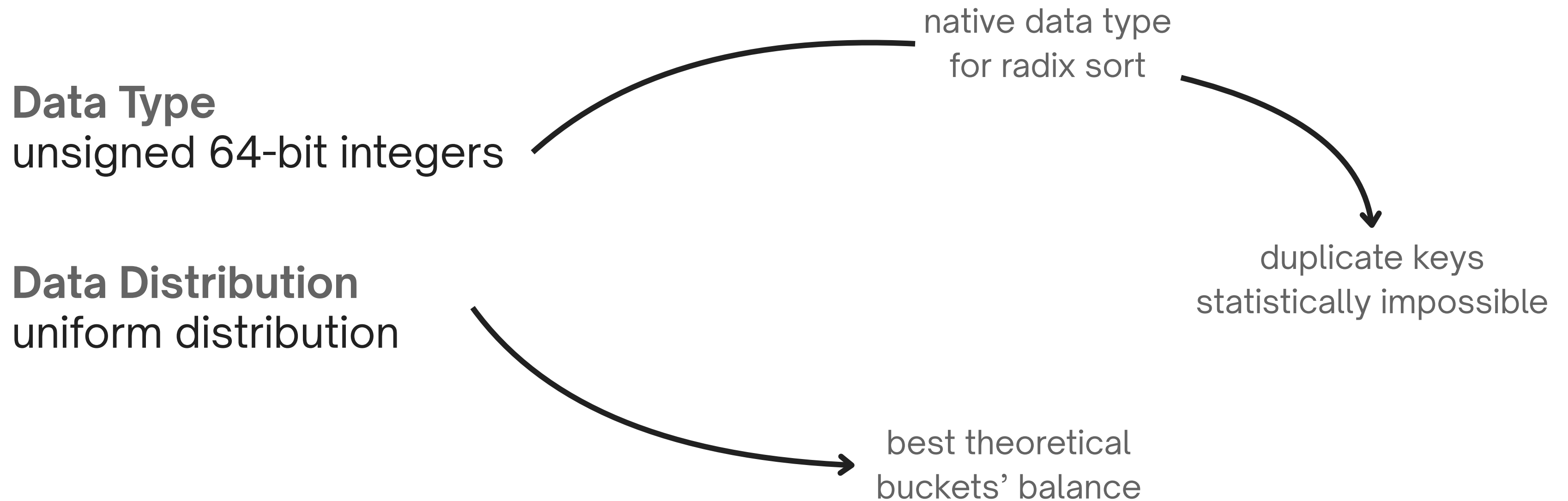
procedure

1. **Local Generation** (N/P keys per process)
2. **Local Sort** (leveraging T threads)
3. **Local Regular Sampling**
4. **Global Pivot Selection** (on process #0)
5. **Local Partitioning** (wrt to the selected pivots)
6. **Global Buckets Exchange** (all to all)
7. **Local K-Way Merge** (one process, one pivot, P buckets)



PARALLEL SAMPLE SORT

assumptions



PARALLEL SAMPLE SORT

local sort

Quick Sort

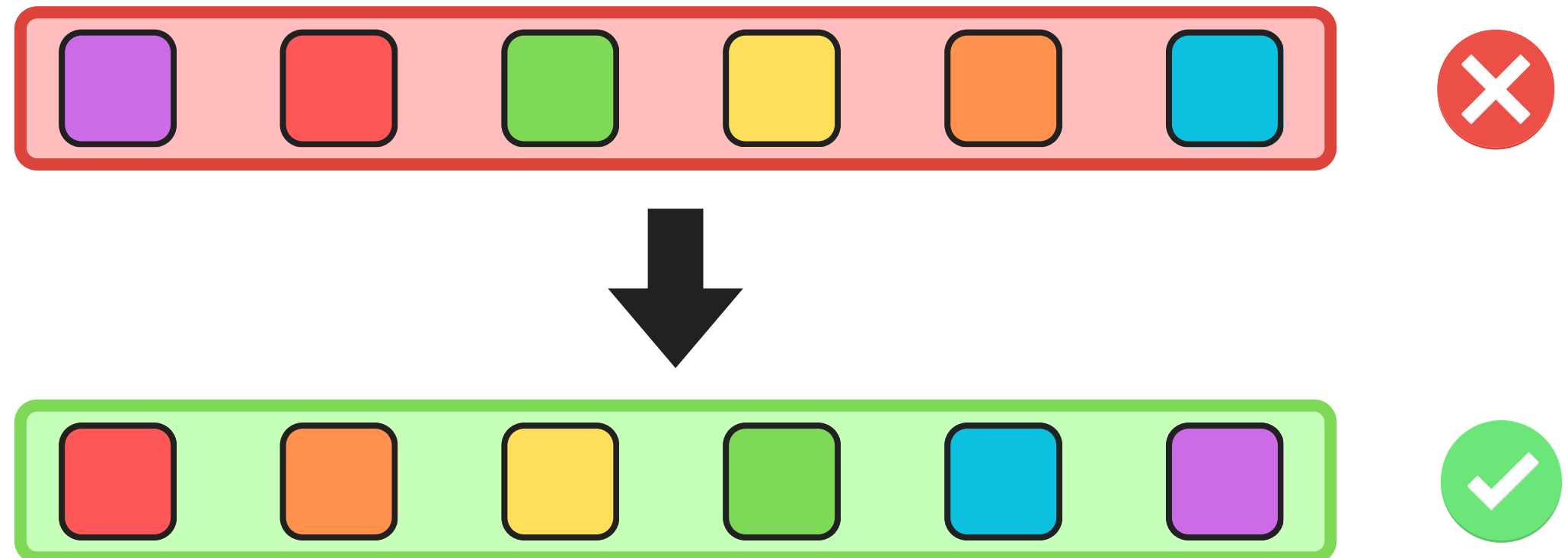
given n keys

$O(n \log(n))$

Radix Sort

given n keys, and digit size d

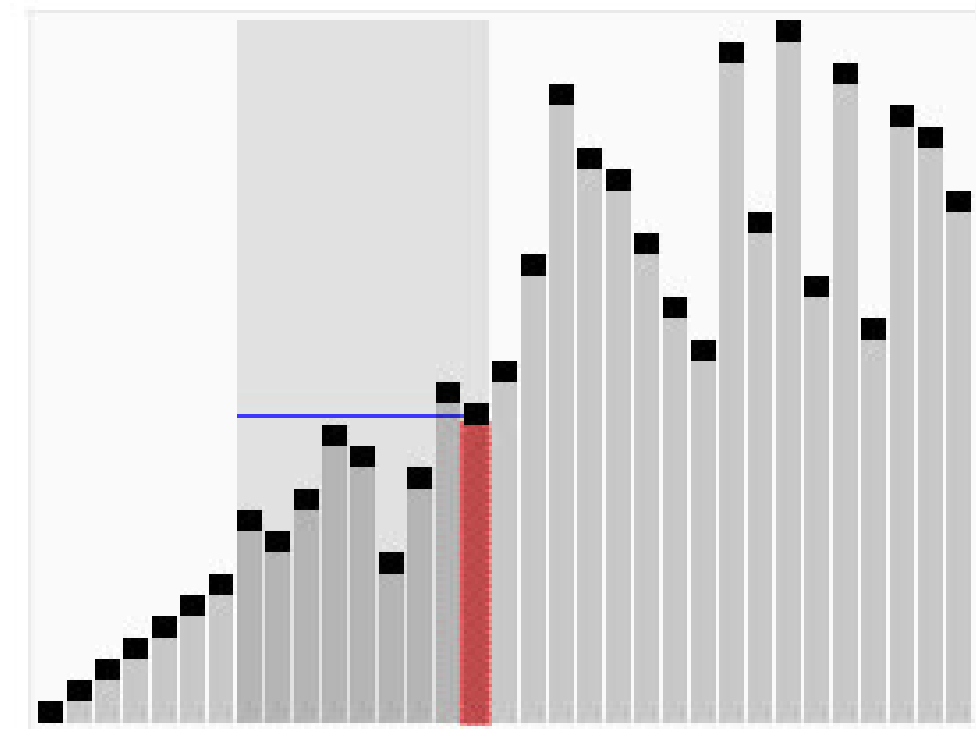
$O(dn)$



QUICK SORT

details

- Recursive Approach
- OMP Tasks
- Serial Cutoff (at 25000 elements)
- Insertion Sort Cutoff (at 8 elements)
- MITM Pivot Selection



source Wikipedia

RADIX SORT

details

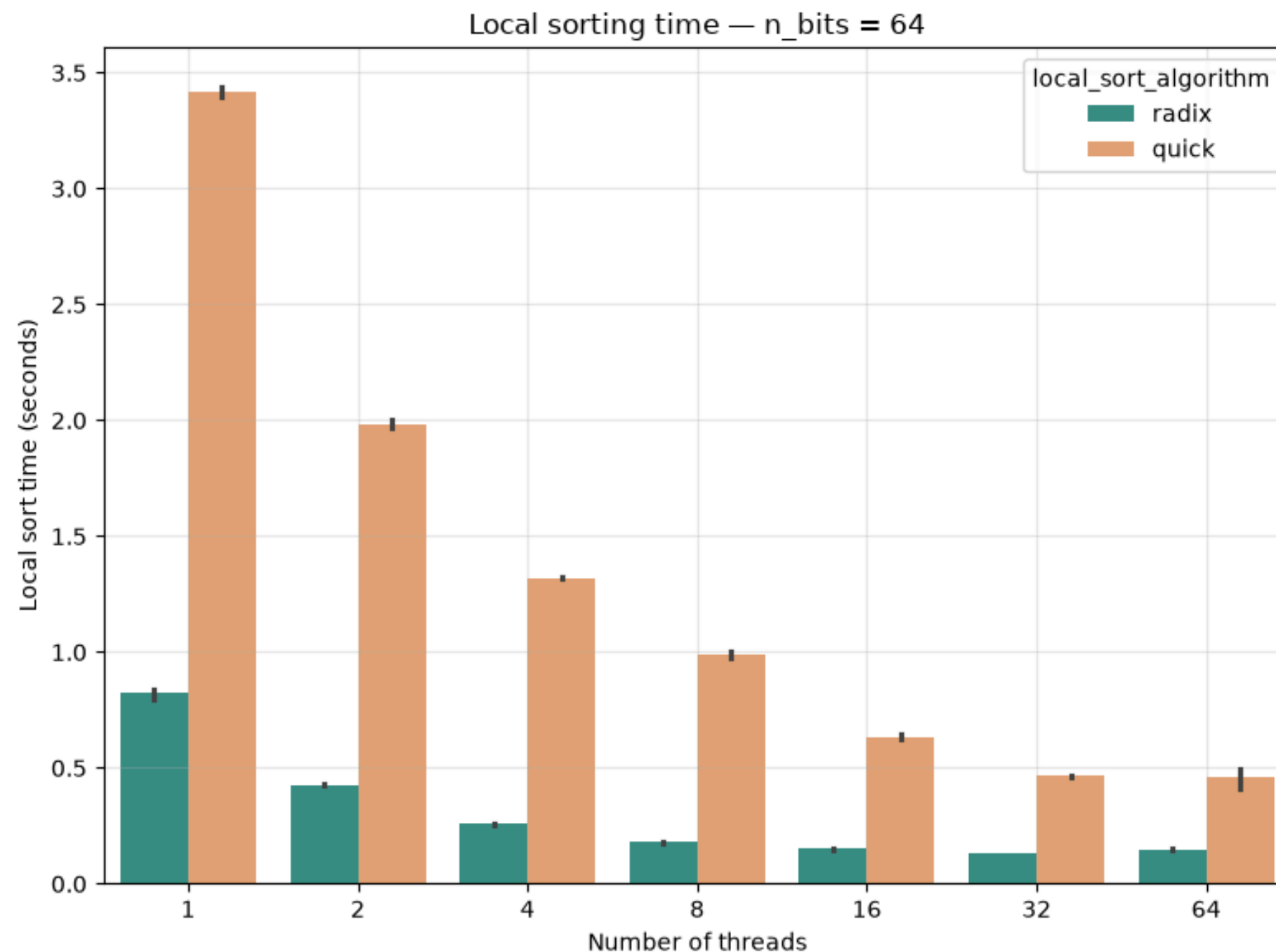
- Stable Counting Sort based
- Digit Sorting Parallelized
- Optimal Digit Size 8

1° PASSO		2° PASSO		3° PASSO		4° PASSO		
253		10		5		5		5
346		253		10		10		10
1034		1034		127		1034		127
10	→	5	→	1034	→	127	→	253
5		346		346		253		346
127		127		253		346		1034

source Wikipedia

PARALLEL SAMPLE SORT

local sort



- nkeys = $5 * 10^7$
- nranks = 1
- distribution = uniform

Radix sort clearly outperform quick sort.

Probable causes:

- radix's ideal configuration
- radix's OMP better implementation

*All other experiments are shown in the report. Refere to it for detailed results. This is meant only as 15 minutes presentation.

PARALLEL SAMPLE SORT

k-way merge

Basic k-way Merge

given n keys, and k data streams

$$O(nk)$$

Bin k-way Merge

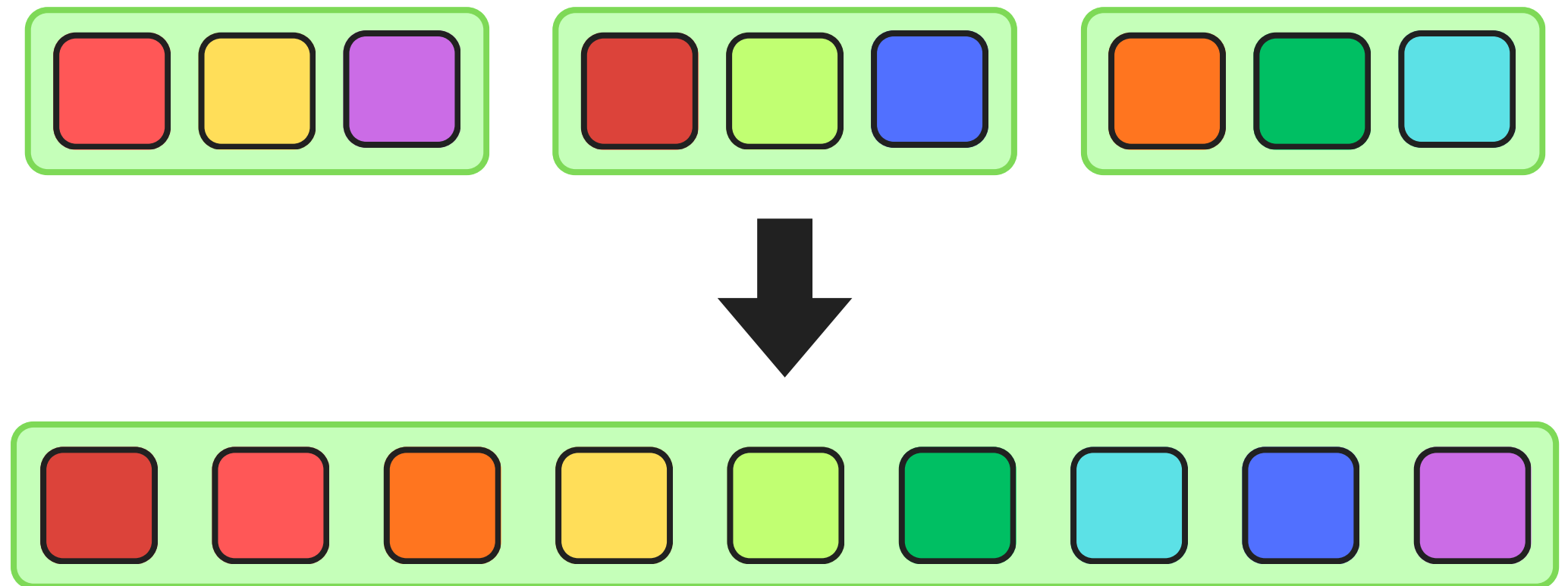
given n keys, and k data streams

$$O(n \log(k))$$

Heap k-way Merge

given n keys, and k data streams

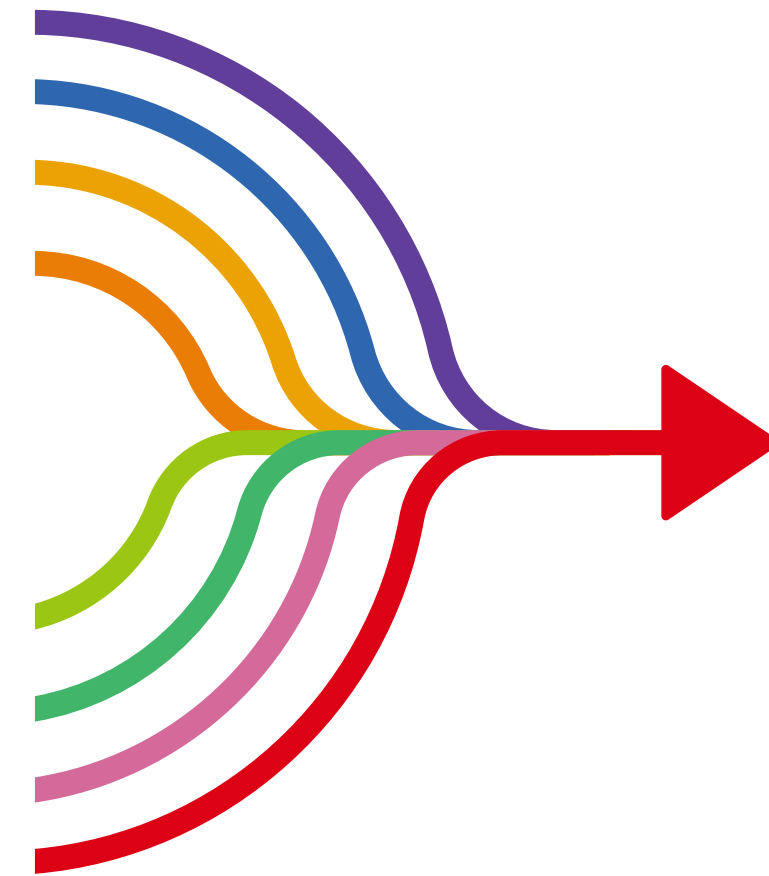
$$O(n \log(k))$$



BASIC K-WAY MERGE

details

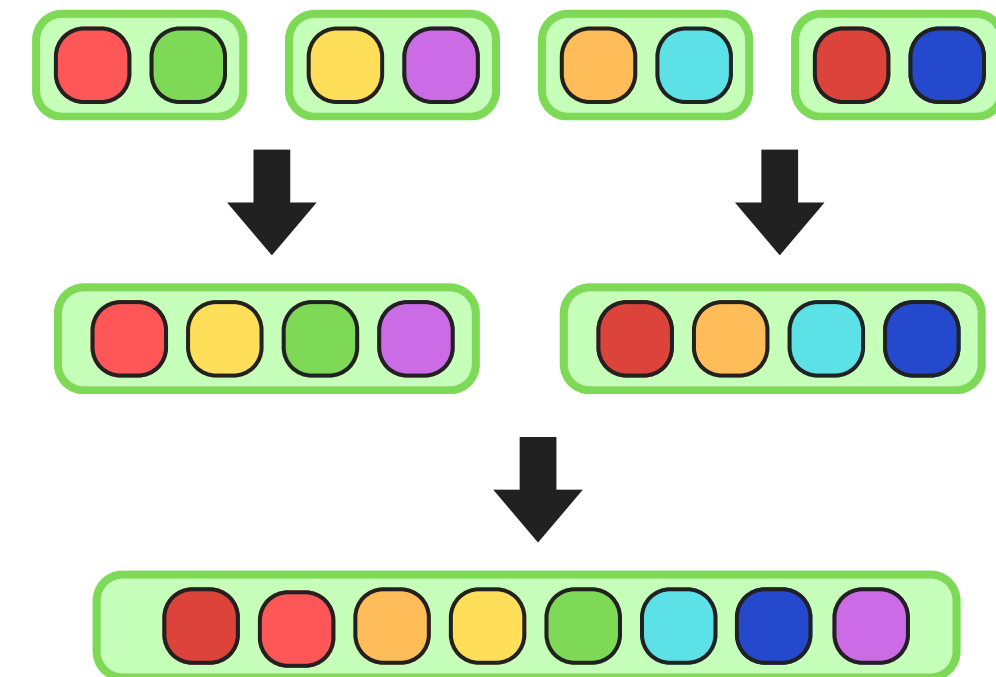
1. Scan the heads of the sorted k input streams
2. Select the minimum among them
3. Add to the output stream
4. Repeat until one stream is left



BINARY K-WAY MERGE

details

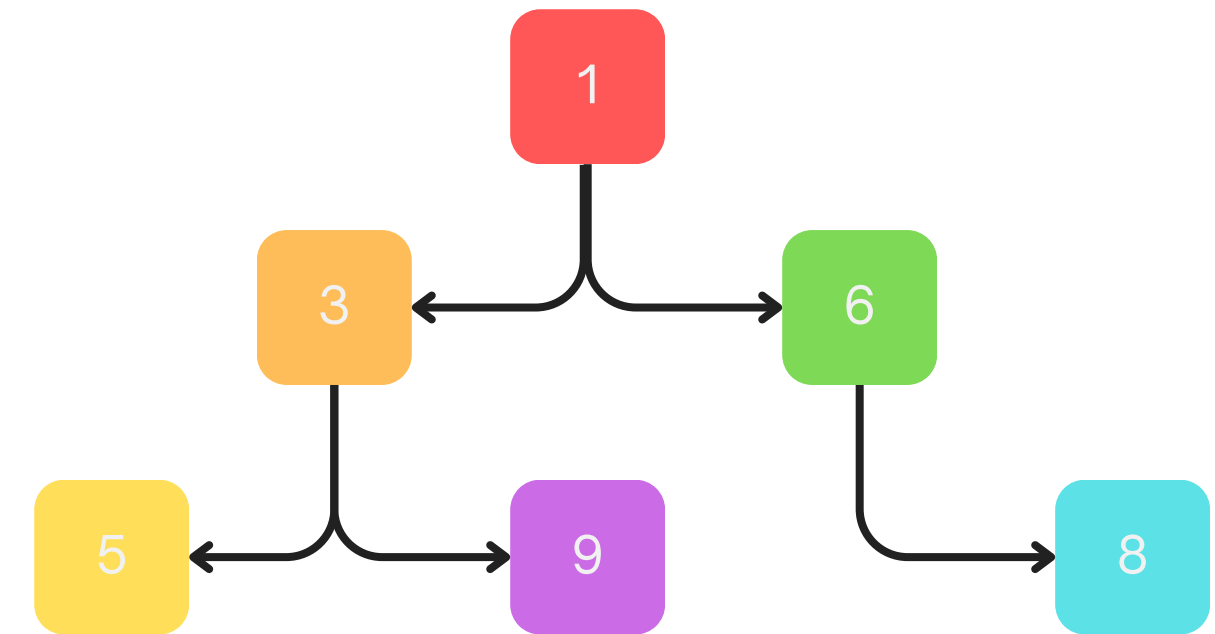
1. Pair input streams (0, 1) (2, 3) ...
2. Use basic 2-way merge on the pairs of streams
3. Repeat until one stream is left (treat the resulting output stream as input streams and pair them again)



HEAP K-WAY MERGE

details

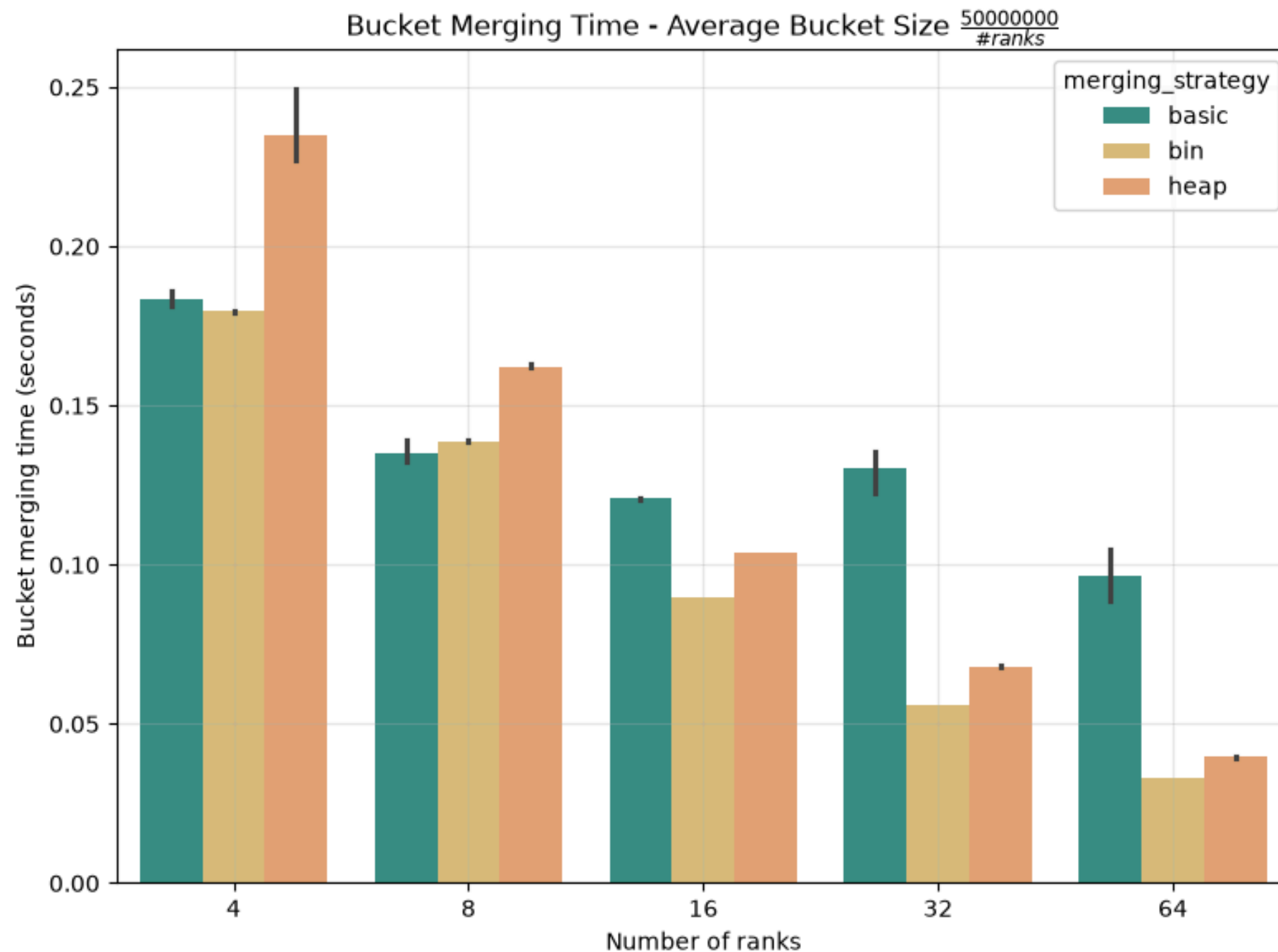
1. Construct a min-heap with the initial elements of each stream
2. Pop the root from the heap and add the following minimum from the stream that contained the root
3. Repeat until the heap is empty



PARALLEL SAMPLE SORT

k-way merge

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- $nkeys = 5 * 10^7$
- $nthreads = 1$
- distribution = uniform

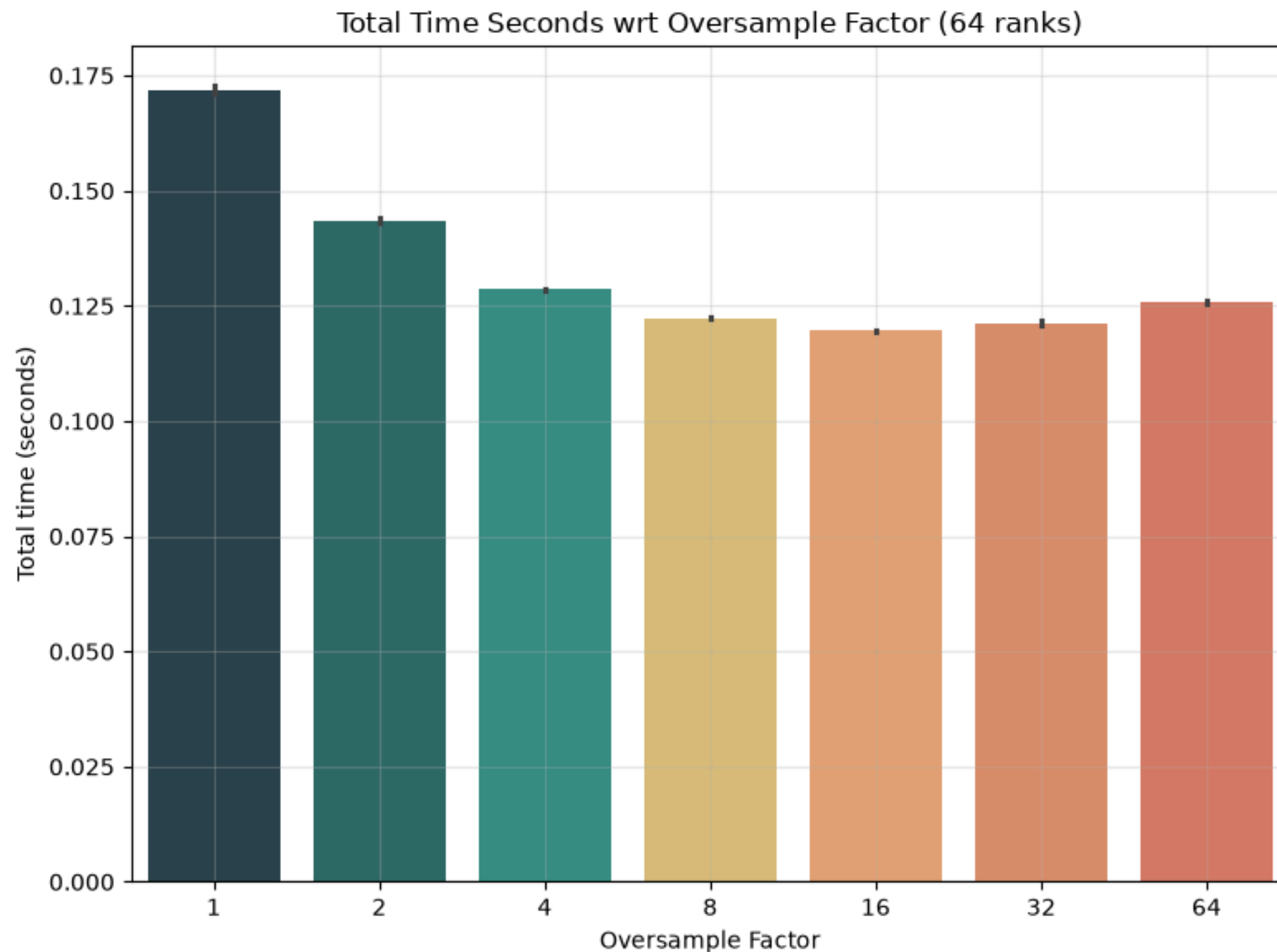
The binary strategy is the most robust among all. It has the same theoretical computational bound of the heap's one but it has less overhead due to no structure maintenance.

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PARALLEL SAMPLE SORT

oversample factor

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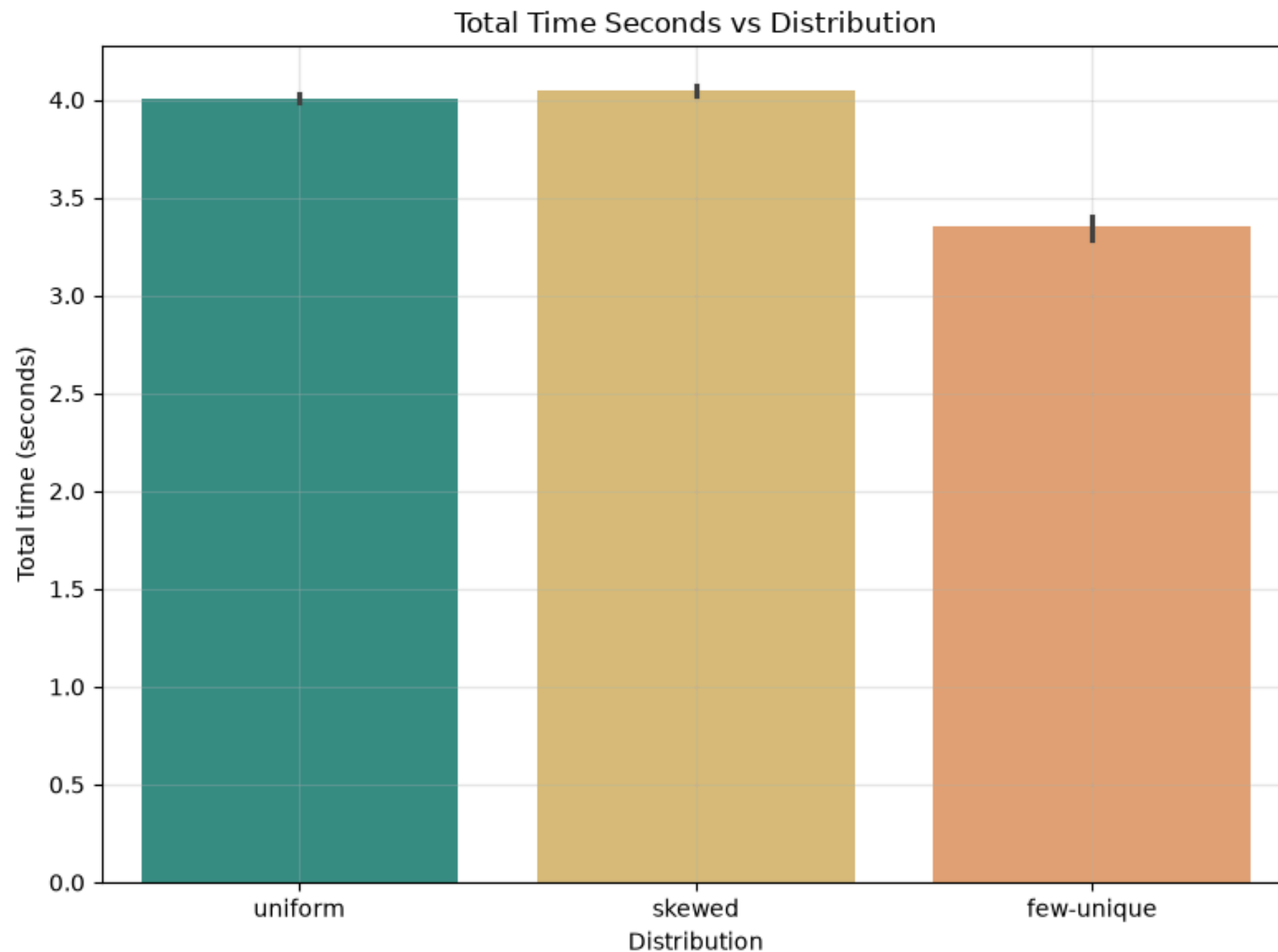
- $nkeys = 5 * 10^7$
- $n ranks = 64$
- $nthreads = 1$
- distribution = uniform

The experiment has been conducted on a similar configuration with 8 ranks and 8 threads per rank. Similar results have been obtained.

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PARALLEL SAMPLE SORT

skewed distribution



- nkeys = 10^9
- nranks = 16
- nthreads = 4
- distribution = uniform
- merging_strategy = bin
- sorting = radix

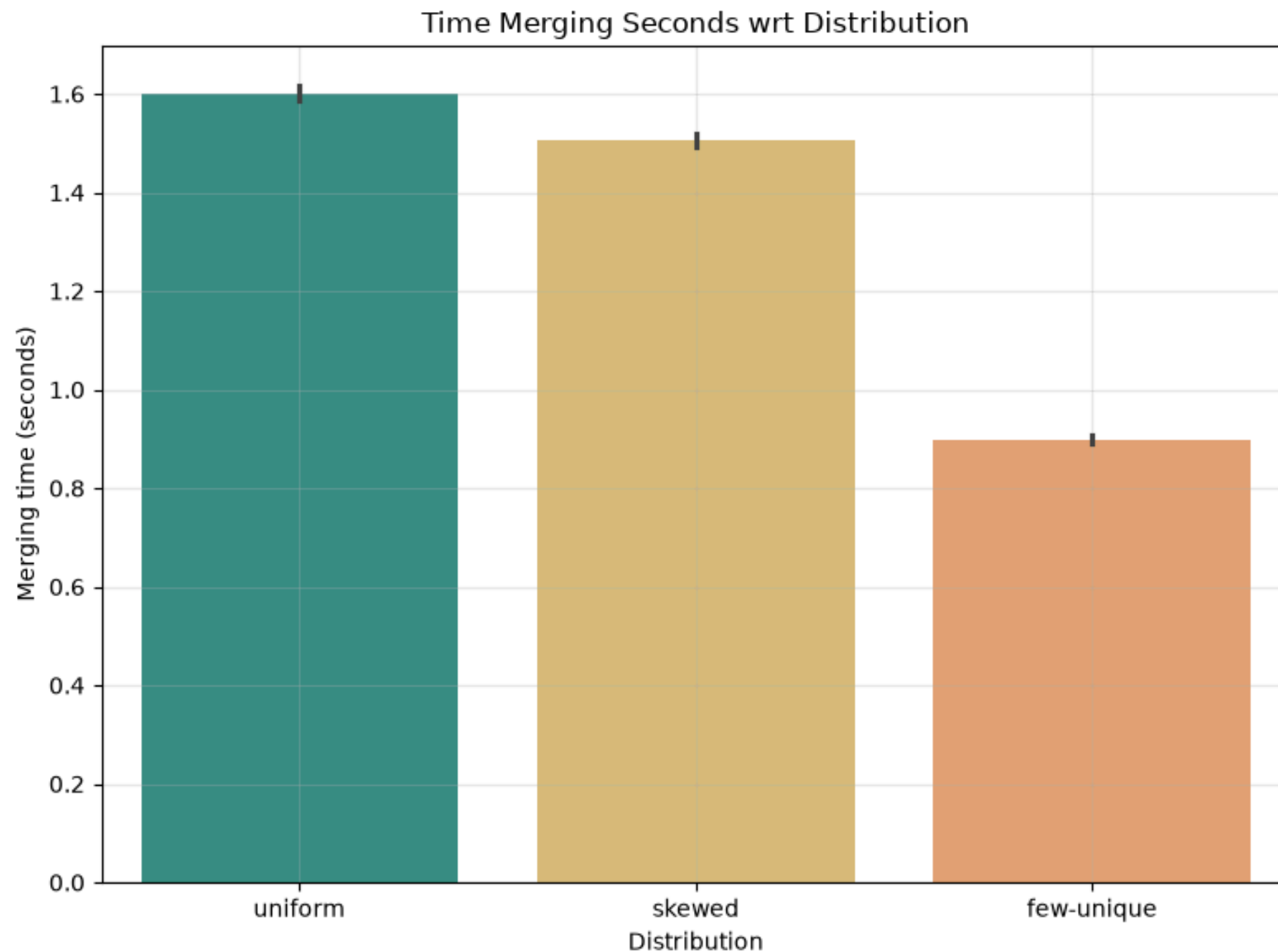
Why this strange results?

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PARALLEL SAMPLE SORT

skewed distribution

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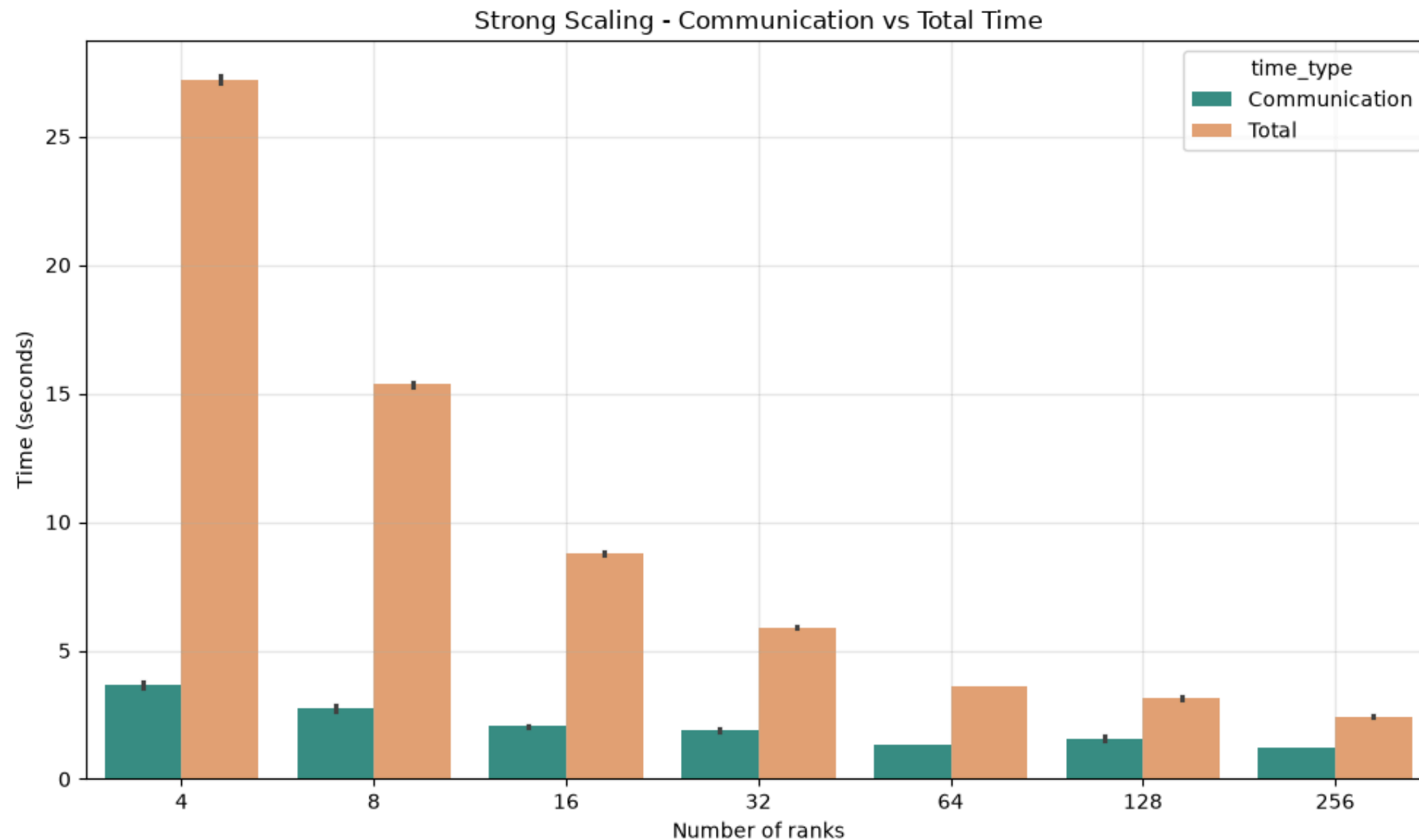
The merging time is the main relevant factor. As a consequence of the non-uniform distribution, a significant number of destination buckets can remain empty. This lead to quicker overall execution.

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STRONG SCALING

total and communication time

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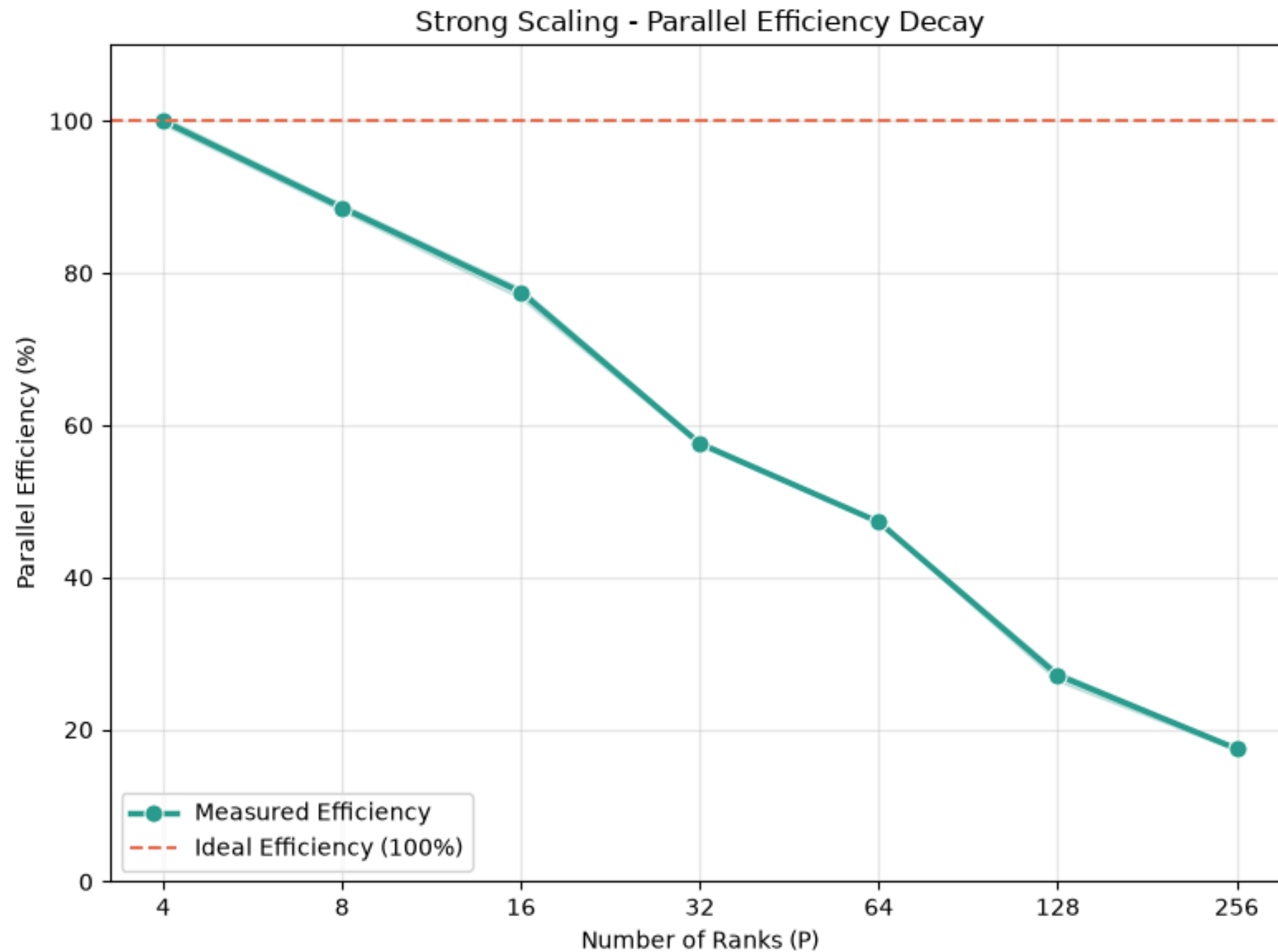
- nkeys = 10^9
- nthreads = 1
- distribution = uniform
- merging_strategy = bin
- sorting = radix

The execution time is high because only one thread is used.

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STRONG SCALING

parallel efficiency decay



$$E_N = \frac{S_N}{N} = \frac{T_1}{NT_N}$$

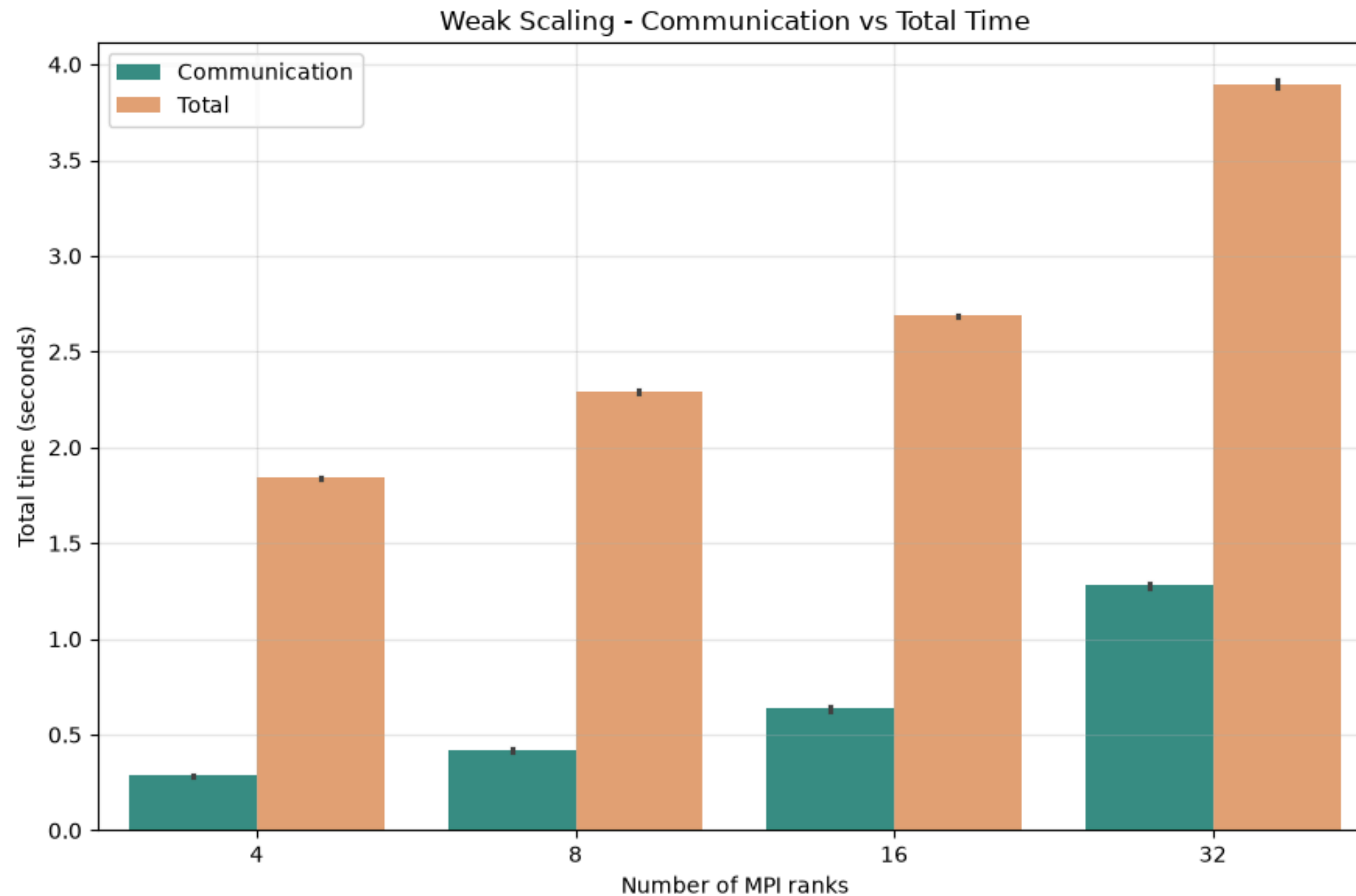
$$E_N^{(4)} = \frac{T_4}{NT_N}$$

$$E_N \approx 4E_N^4$$

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WEAK SCALING

total and communication time

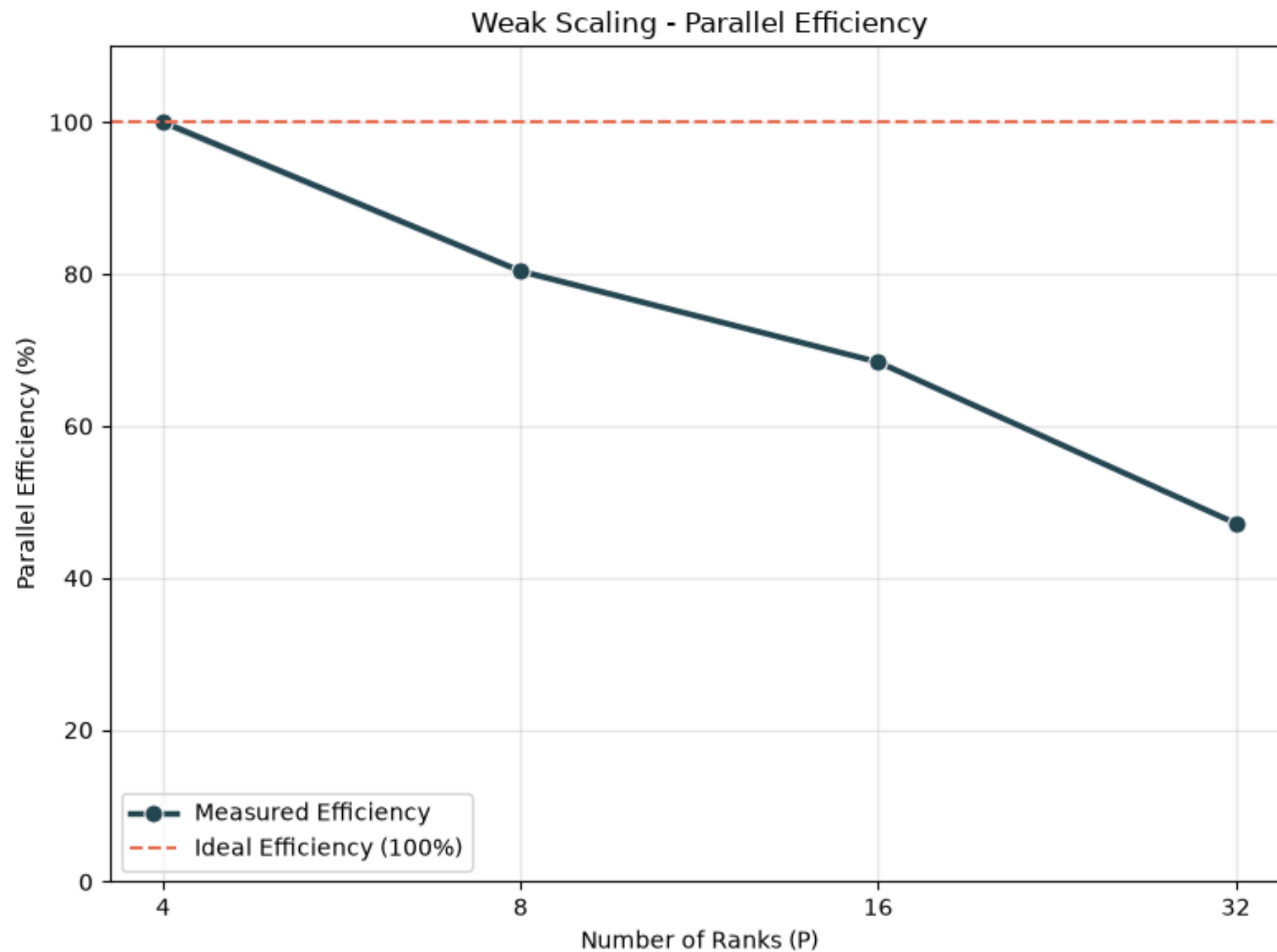


- $nkeys/nranks = 2 \times 10^7$
- $nthreads = 4$
- $distribution = uniform$
- $merging_strategy = bin$
- $sorting = radix$

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WEAK SCALING

parallel efficiency decay

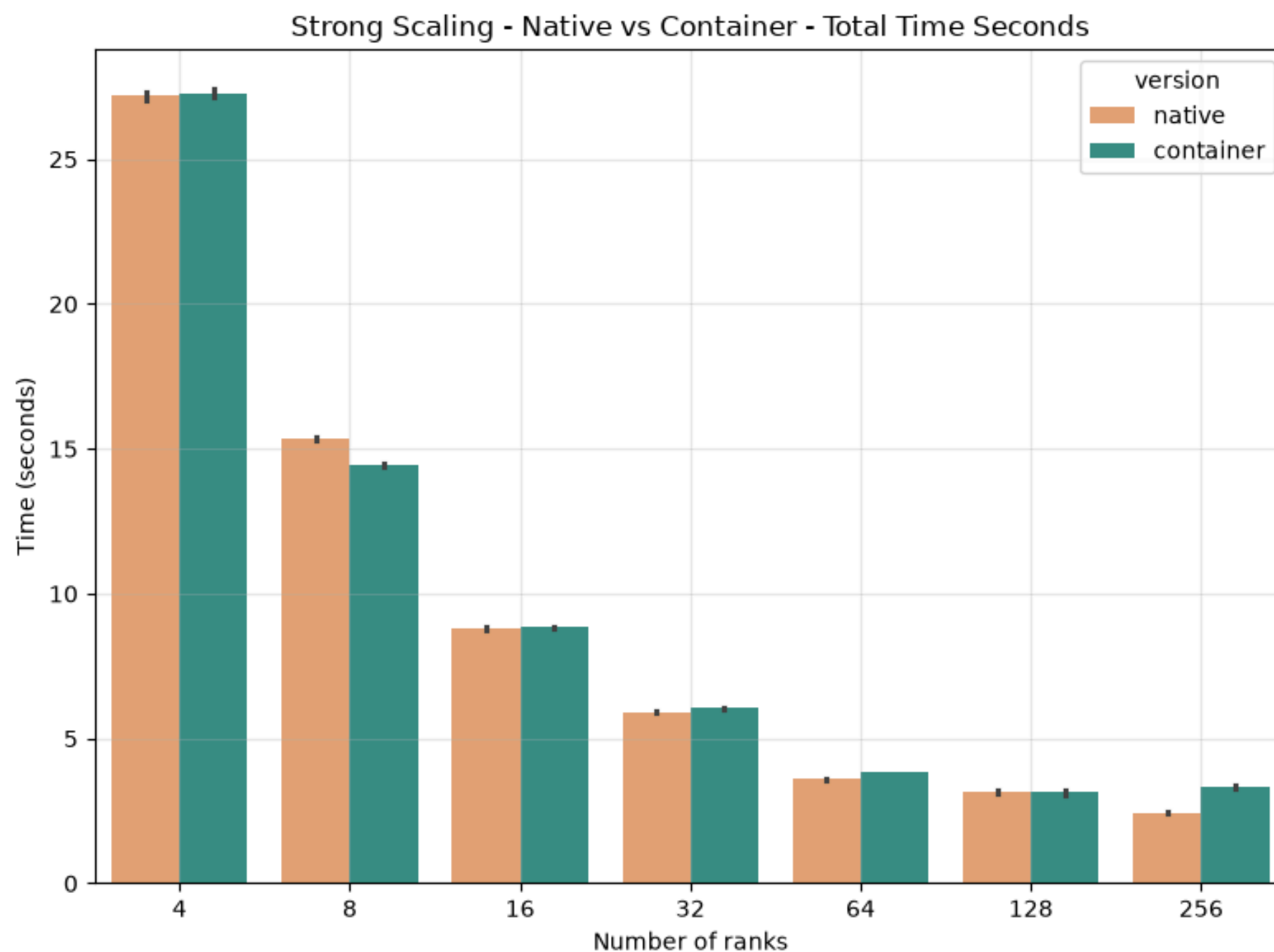


$$E_N \approx \frac{T_4}{T_N} \times 100$$

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NATIVE - CONTAINER

strong scaling



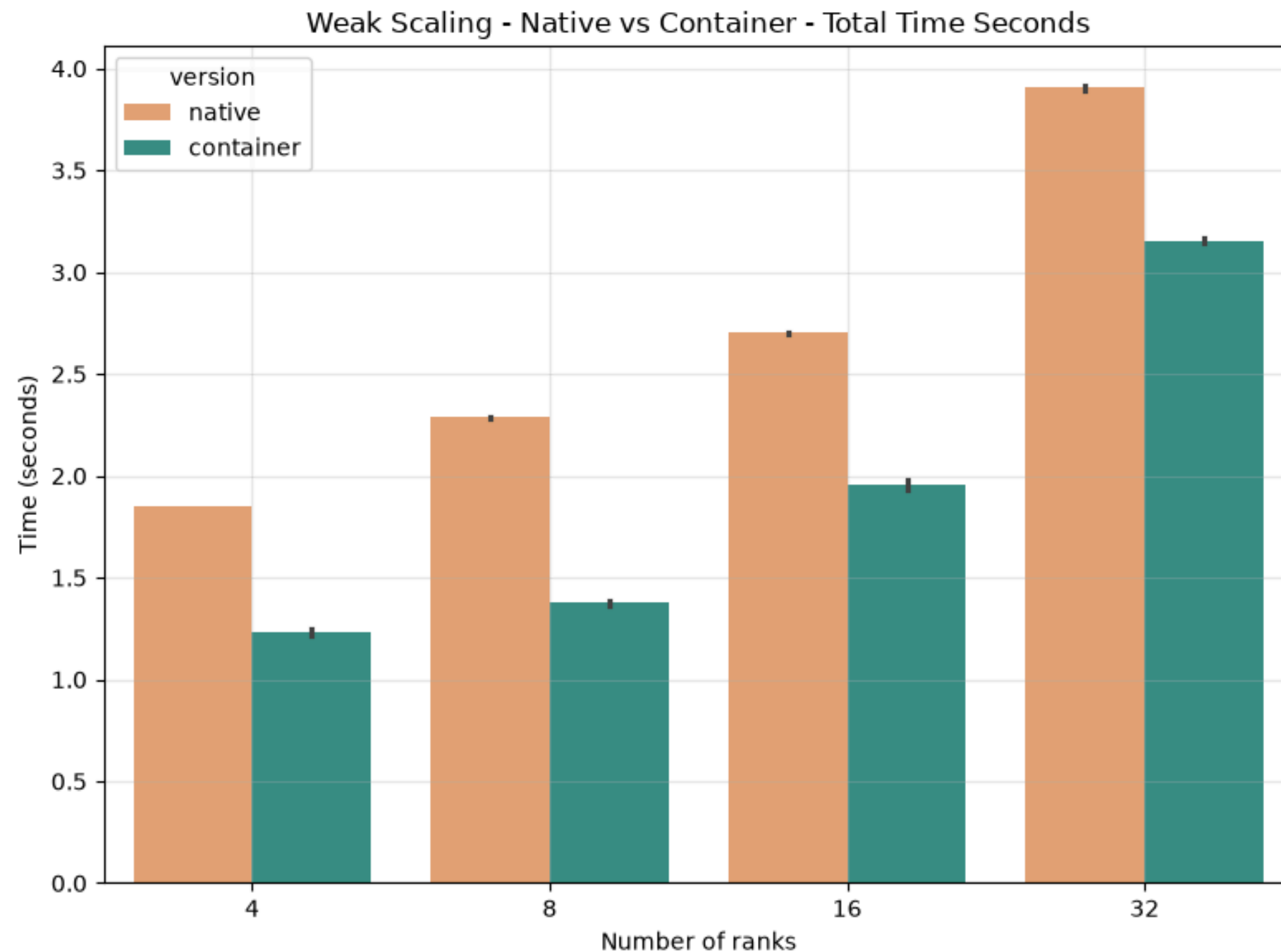
This is a nice comparable behavior. Looking at the other timings seems like everything is pretty balanced across the two different implementations. Notice, there is no OMP parallelization going on here.

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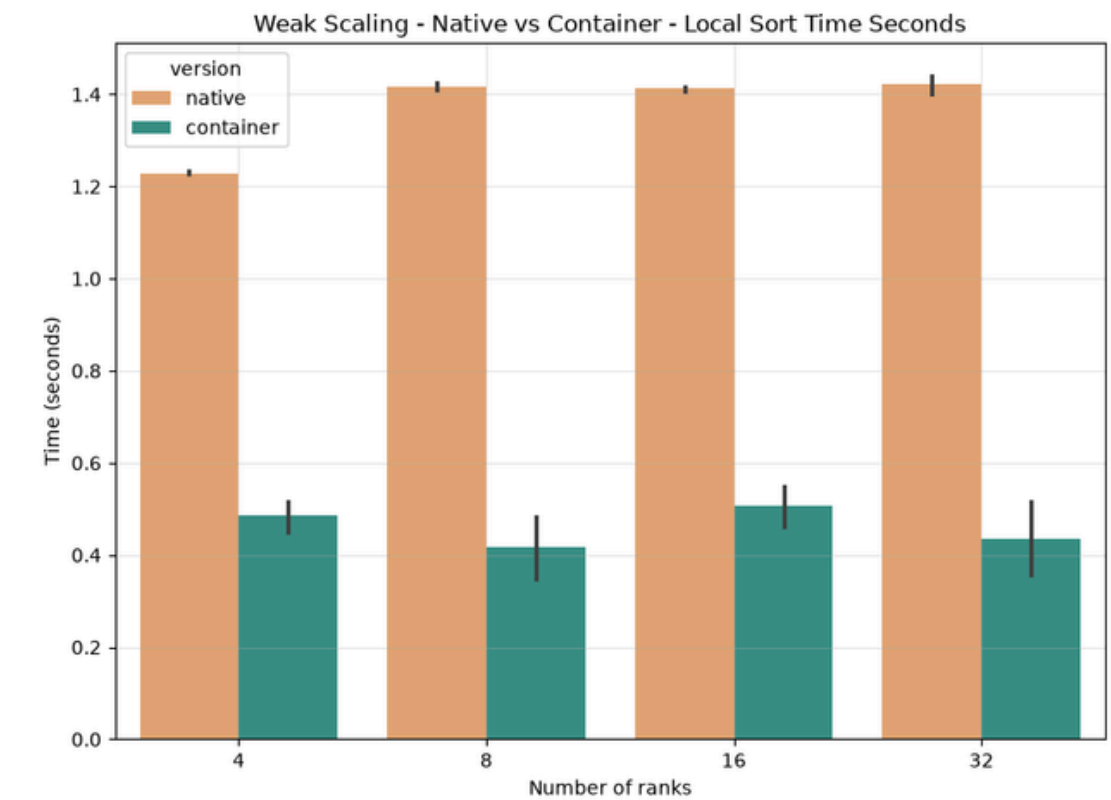
NATIVE - CONTAINER

weak scaling

22



this is almost surely caused by
different OMP versions

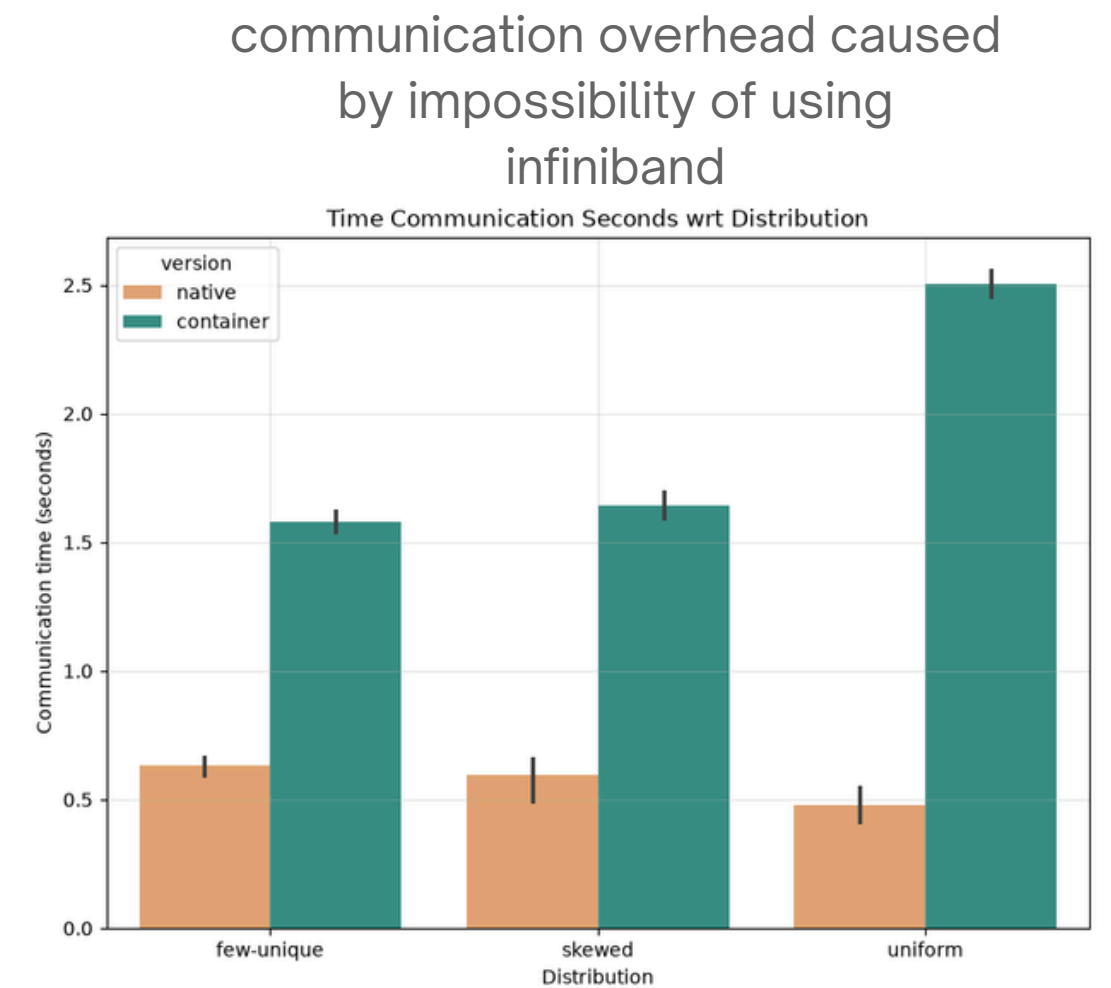
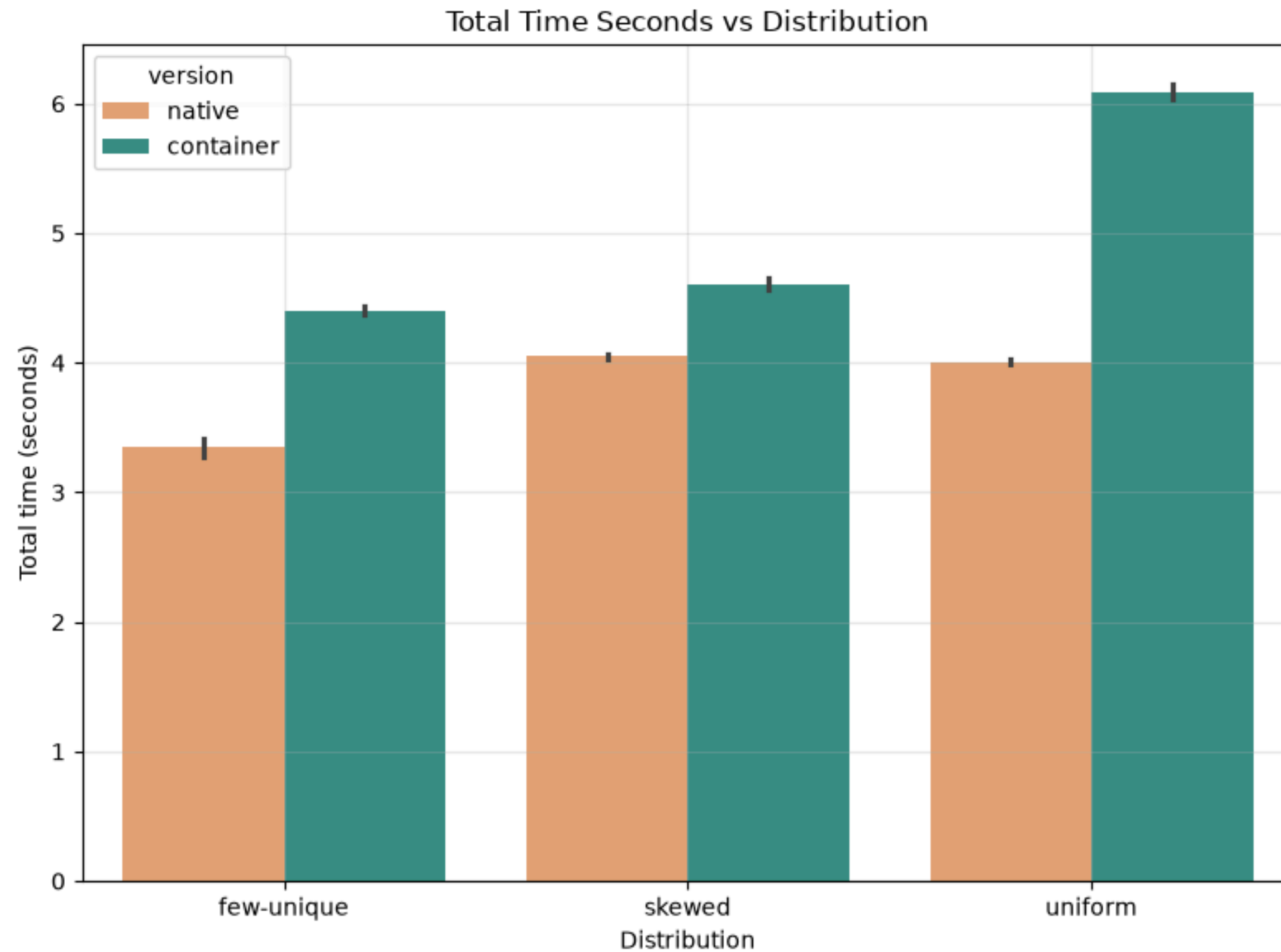


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CONTAINER - NATIVE

skewed distribution

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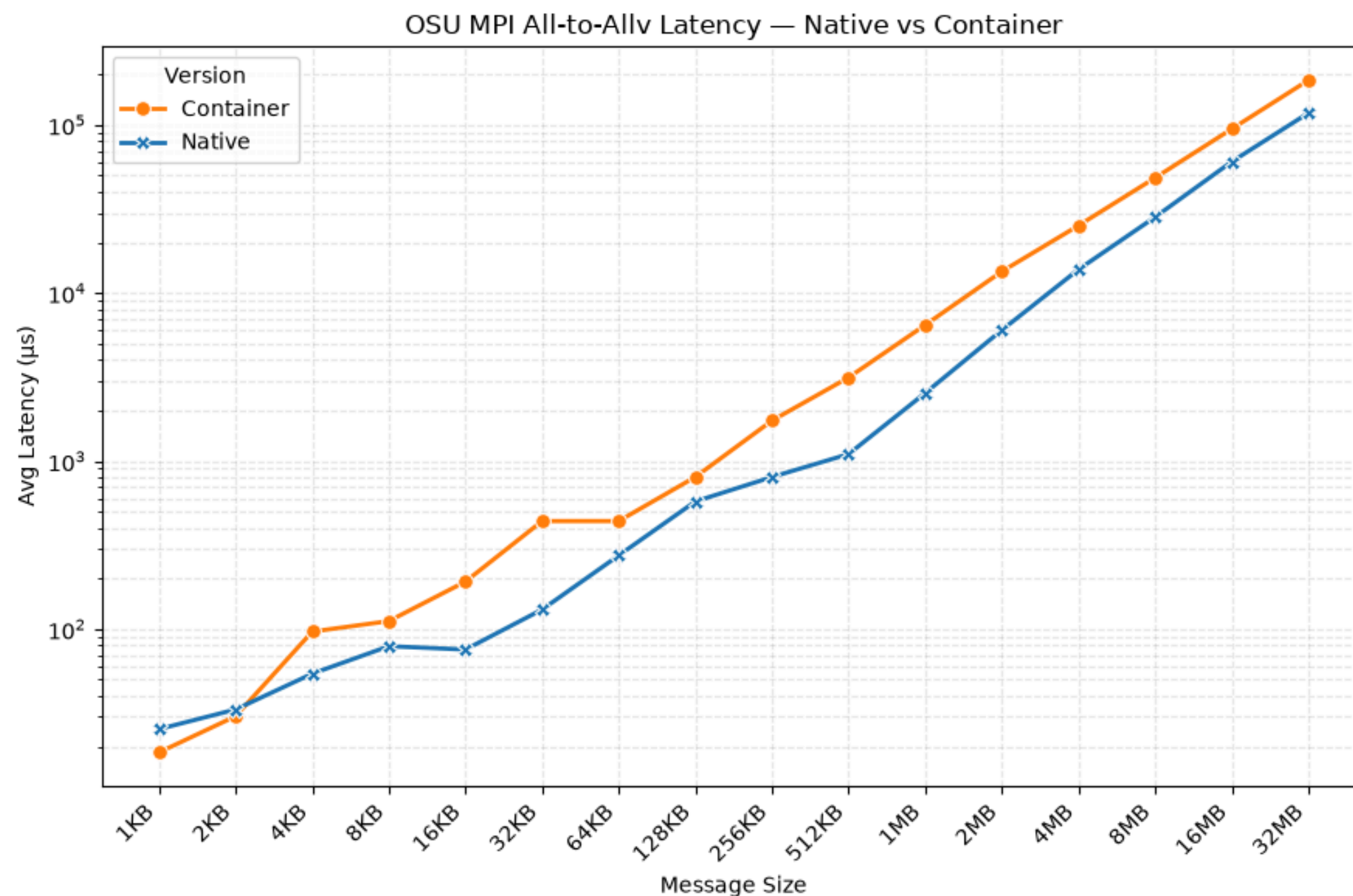


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OSU MICROBENCHMARK

OSU Alltoallv

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This confirm the communication overhead observed in the various experiments. This cannot tells anything about the decreased execution time.

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**thank you for the
attention**